Graphs :

: Modulus form

: Greatest integer function [GIF]

(or)

Step Function

: Fractional Part

: $\log x$

: Equations :

→ Quadratic

→ Cubic

→ n^{th} degree Polynomial

: Trigonometric Functions, inverse Trigonometric functions

: increasing, decreasing, Stricty increasing
Stricty decreasing !

** Slope May equal to Zero : increase decreasing

Slope can't be equal to zero : Stricty increasing
Stricty decreasing.

: Exponential functions

: Signum function. : "Cases in Quadratic"

** $\boxed{\{x\} = x - [x]}$

Modulus Function :

Case - 1 :

$y = |f(x)|$

⇨ Take reflection Over X ...

Case - 2 :

$y = f(|x|)$

⇨ Take reflection Over Y

Left right :

$y = f(x)$

$y = f(x+a)$

- Graph Shifts towards left by a units ($a>0$)
- Graph Shifts towards right by a Units ($a<0$)

Up and down :

$y = f(x)$

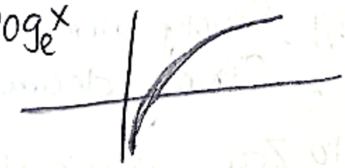
$y = f(x)+a$

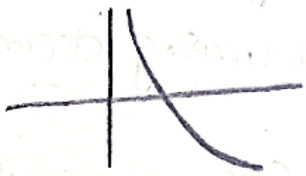
- Graph Shifts Upwards by a Units ($a>0$)
- Graph Shifts downwards by a Units ($a<0$).

: $\log x$:

Case 1

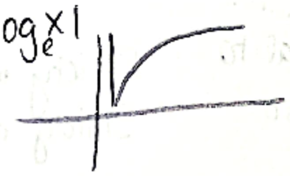
$y = \log_e x$





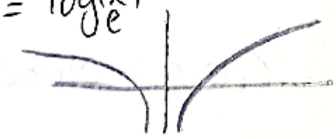
Case 2

$y = |\log_e x|$



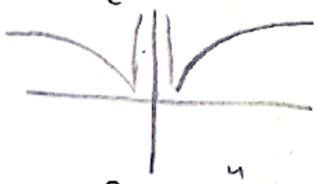
Case-3

$y = \log_e |x|$

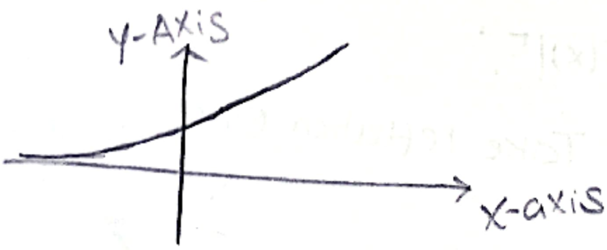


Case-4

$y = |\log_e |x||$



"Exponential function" $y = f(x) = a^x$ ($a>1$)



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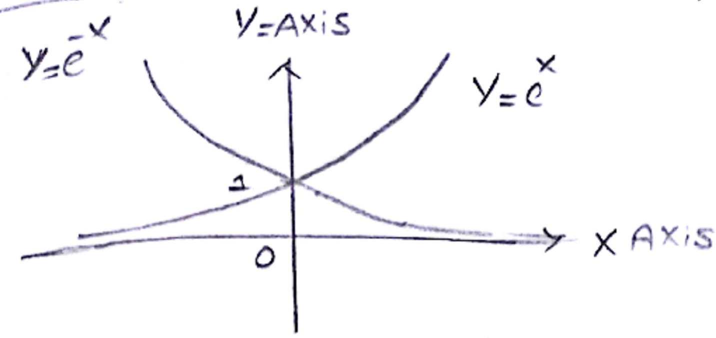
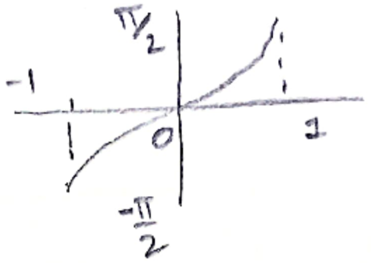
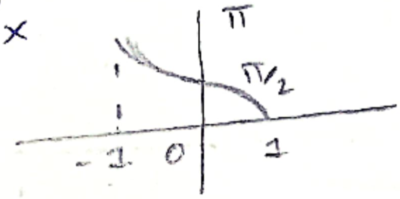


- Inverse Trigonometric function :

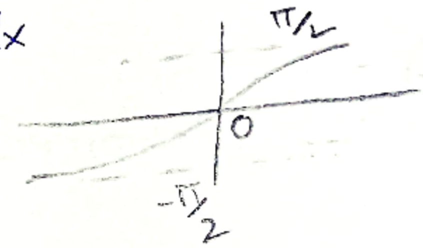
$y = \sin^{-1}(x)$



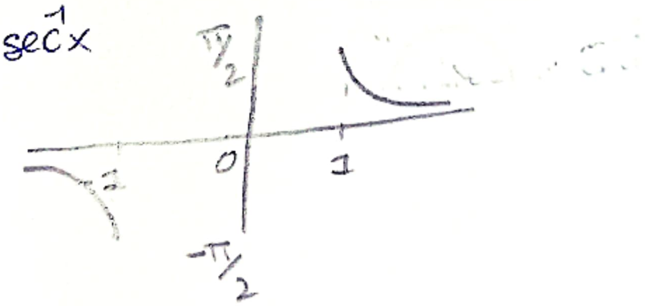
$y = \cos^{-1}x$



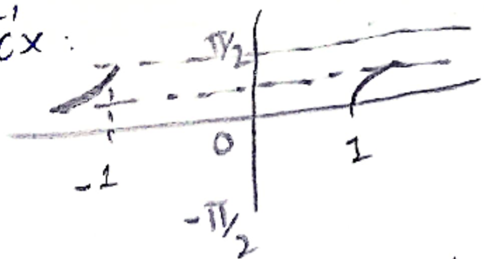
$y = \tan^{-1}x$



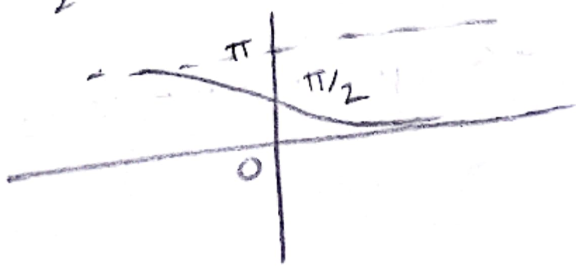
$y = \operatorname{cosec}^{-1}x$



$y = \sec^{-1}x$:



$y = \cot^{-1}x$



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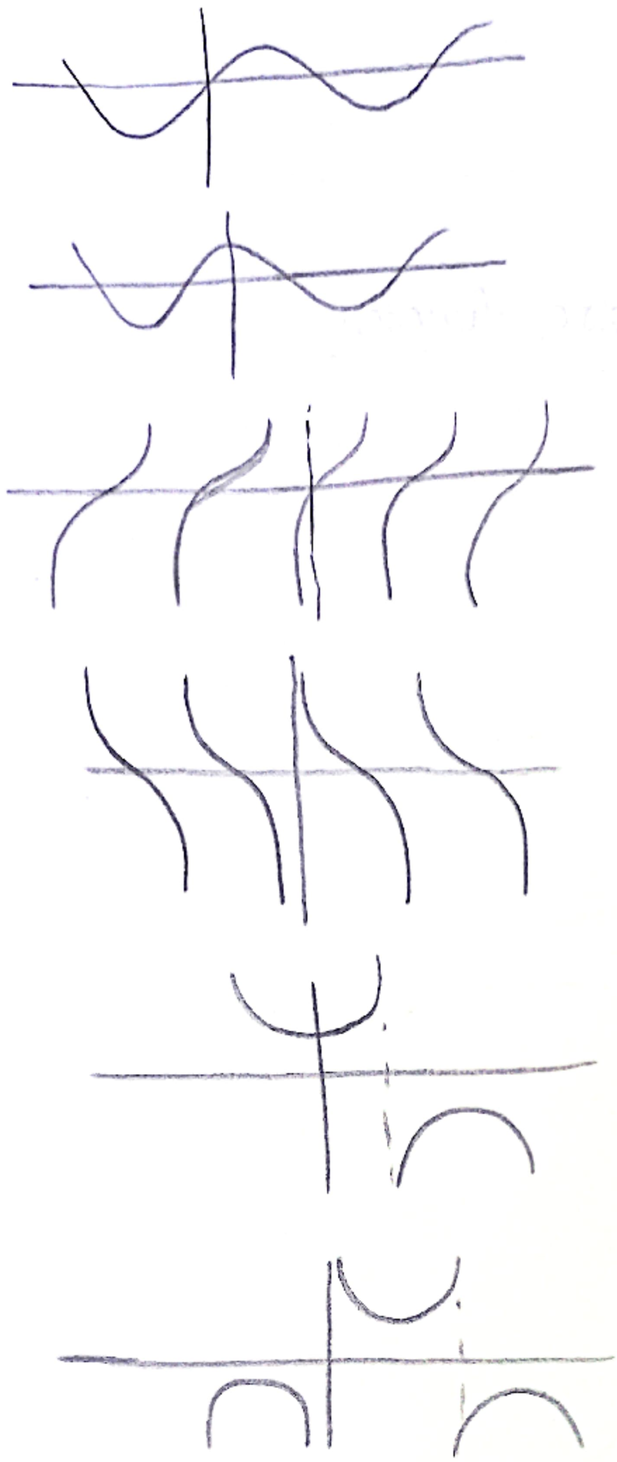
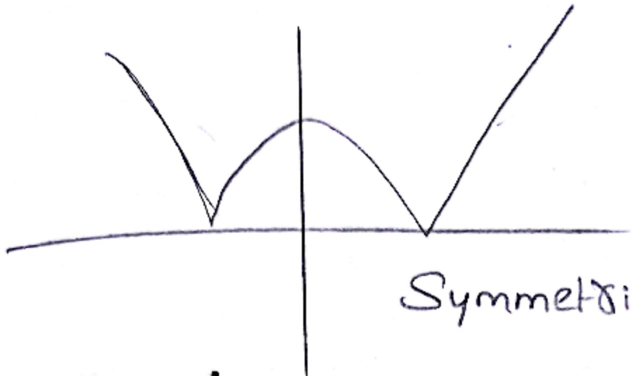


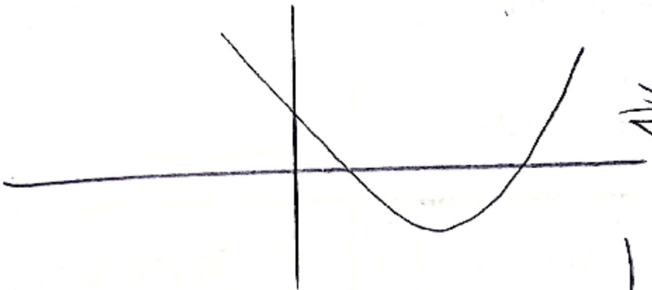
Quadratic Equation :

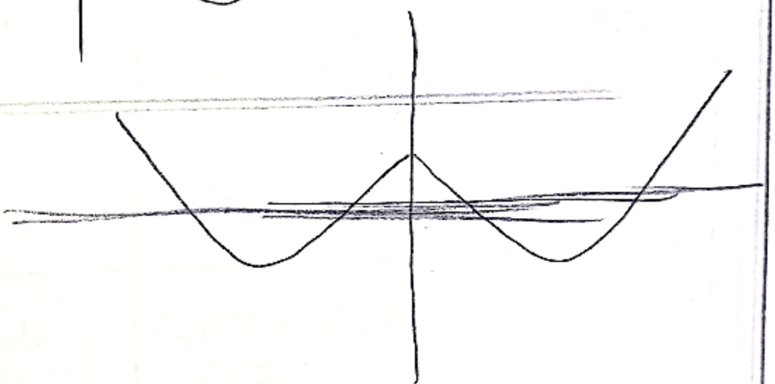
- Case 1 : $y = |ax^2+bx+c|$



Symmetric about X axis...

- Case 2 : $y = ax^2+b|x|+c$





Symmetric About Y axis...